

Inverse Problems in Geophysics

Part 4: Singular Value Decomposition

2. MGPY+MGIN

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Recap

Over-determined problems ($N \geq M$)

- least-squares solution model-sided (inverse of $\mathbf{G}^T \mathbf{G}$)
- perfect model resolution (unique model), imperfect data

Under-determined problems ($N < M$)

- least-norm solution data sided (inverse of $\mathbf{G} \mathbf{G}^T$)
- perfect data resolution (all data perfectly fit), imperfect model

The Singular Value Decomposition (SVD)

Splitting an inverse problem into the fundamental ingredients

The Singular Value Decomposition

Any matrix $\mathbf{A}$ can be decomposed into model and data eigenvectors, weighted by singular values:

$$\mathbf{A} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \cdot \mathbf{v}_i^T = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$$

with the eigenvalues in $\mathbf{\Sigma} = \text{diag}(\sigma_i)$, a set of orthogonal data eigenvectors $\mathbf{U} \in \mathbb{R}^{N \times N}$ with $\mathbf{U}^{-1} = \mathbf{U}^T$, and model eigenvectors $\mathbf{V} \in \mathbb{R}^{M \times M}$ with $\mathbf{V}^{-1} = \mathbf{V}^T$.

Derivation - eigenvalue decomposition

The quadratic matrix $\mathbf{A}$ projects a vector in another direction. Special vectors are eigenvectors who keep their direction

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$$

The solution of the equation $\mathbf{A} - \lambda\mathbf{I} = 0$ leads to eigenvalues λ over the characteristic polynome $\det(\mathbf{A} - \lambda\mathbf{I})$ that correspond to eigenvectors in the matrix $\mathbf{Q}$ by

$$\mathbf{A} = \mathbf{Q}\text{diag}(\lambda_i)\mathbf{Q}^T = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$$

Derivation (2)

We create a symmetric matrix from $\mathbf{G}$ and its transpose

$$\mathbf{A} = \begin{pmatrix} 0 & \mathbf{G} \\ \mathbf{G}^T & 0 \end{pmatrix}$$

It has the eigenvalue decomposition (two coupled eigenvalue problems)

$$\mathbf{A}\mathbf{x}_i = \lambda_i\mathbf{x}_i \quad \text{with} \quad \mathbf{x}_i = \begin{pmatrix} \mathbf{u}_i \\ \mathbf{v}_i \end{pmatrix}$$

$$\Rightarrow \mathbf{G}\mathbf{v} = \lambda\mathbf{u} \quad \text{and} \quad \mathbf{G}^T\mathbf{u} = \lambda\mathbf{v}$$

Derivation (3)

$$\Rightarrow \mathbf{G}\mathbf{v} = \lambda\mathbf{u} \quad \text{and} \quad \mathbf{G}^T\mathbf{u} = \lambda\mathbf{v}$$

Through multiplication with $\mathbf{G}^T$ or $\mathbf{G}$ we obtain

$$\mathbf{G}^T\mathbf{G}\mathbf{v} = \lambda^2\mathbf{v} \quad \text{and} \quad \mathbf{G}\mathbf{G}^T\mathbf{u} = \lambda^2\mathbf{u}$$

i.e. two eigenvalue problems for model space ($\mathbf{v}$) and data space ($\mathbf{u}$)

$$\mathbf{G} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T \quad \text{and} \quad \mathbf{G}^T = \mathbf{V}\mathbf{\Sigma}\mathbf{U}^T \quad \text{with} \quad \mathbf{\Sigma} = \text{diag}(\lambda_i)$$

Model and Data space

$\mathbf{U} \in \mathbb{R}^{N \times N}$ spans the data space, $\mathbf{V} \in \mathbb{R}^{M \times M}$ spans the model space

the belong together with a number r (rank) of non-zero singular values

$\mathbf{U}$ and $\mathbf{V}$ can be split into a data/model space and a null space

$$\mathbf{U} = [\mathbf{U}_r, \mathbf{U}_0]$$

$$\mathbf{V} = [\mathbf{V}_r, \mathbf{V}_0]$$

Eigenvectors and singular values

The eigenvectors associated with zero singular values ($\sigma_i = 0$) span the null spaces of data and model. The number of non-zero eigenvalues r defines the rank of the matrix which can be abbreviated by

$$\mathbf{A}_r = \sum_{i=1}^r \sigma_i \mathbf{u}_i \cdot \mathbf{v}_i^T = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^T$$

where $\mathbf{B}_r$ holds the first r columns of the matrix $\mathbf{B}$.

A matrix can be approximated by choosing the (pseudo)rank by hand.

Operators in terms of SVD

The forward operator $\mathbf{G}\mathbf{m}$ and its adjoint $\mathbf{G}^T$ are written:

$$\mathbf{G} = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^T$$

$$\mathbf{G}^T = \mathbf{V}_r \mathbf{\Sigma}_r \mathbf{U}_r^T$$

$$\mathbf{G}^T \mathbf{G} = \mathbf{V}_r^T \mathbf{\Sigma}_r \mathbf{U}_r^T \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r = \mathbf{V}_r^T \mathbf{\Sigma}_r^2 \mathbf{V}_r$$

$$\mathbf{G} \mathbf{G}^T = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r \mathbf{V}_r^T \mathbf{\Sigma}_r \mathbf{U}_r^T = \mathbf{U}_r \mathbf{\Sigma}_r^2 \mathbf{U}_r^T$$

The generalized inverse

$$\mathbf{G}_r \mathbf{m} = \mathbf{U}_r \boldsymbol{\Sigma}_r \mathbf{V}_r^T \mathbf{m} = \mathbf{d}$$

$$\mathbf{U}_r^T \mathbf{U}_r \boldsymbol{\Sigma}_r \mathbf{V}_r^T \mathbf{m} = \boldsymbol{\Sigma}_r \mathbf{V}_r^T \mathbf{m} = \mathbf{U}_r^T \mathbf{d}$$

$$\boldsymbol{\Sigma}_r^{-1} \boldsymbol{\Sigma}_r \mathbf{V}_r^T \mathbf{m} = \mathbf{V}_r^T \mathbf{m} = \boldsymbol{\Sigma}_r^{-1} \mathbf{U}_r^T \mathbf{d}$$

$$\mathbf{V}_r \mathbf{V}_r^T \mathbf{m} = \mathbf{m} = \mathbf{V}_r \boldsymbol{\Sigma}_r^{-1} \mathbf{U}_r^T \mathbf{d}$$

$$\Rightarrow \mathbf{m} = \mathbf{G}^\dagger \mathbf{d} \quad \text{with} \quad \mathbf{G}^\dagger = \mathbf{V}_r \boldsymbol{\Sigma}_r^{-1} \mathbf{U}_r^T$$

The SVD provides a generalized (Moore-Penrose) inverse

Resolution

The model resolution

$$\mathbf{R}^M = \mathbf{G}^\dagger \mathbf{G} = \mathbf{V}_p \boldsymbol{\Sigma}_p^{-1} \mathbf{U}_p^T \mathbf{U}_p \boldsymbol{\Sigma}_p \mathbf{V}_p^T = \mathbf{V}_p \mathbf{V}_p^T$$

depends only on the model eigenvectors!

And the data resolution

$$\mathbf{R}^D = \mathbf{G} \mathbf{G}^\dagger = \mathbf{U}_p \boldsymbol{\Sigma}_p \mathbf{V}_p^T \mathbf{V}_p \boldsymbol{\Sigma}_p^{-1} \mathbf{U}_p^T = \mathbf{U}_p \mathbf{U}_p^T$$

depends only on the data eigenvectors!

Inverse problem types - classification scheme

The rank r determines the type of the inverse problem

 **Even-determined**

$$M = N = r$$

 **Over-determined**

$$N > r \geq M$$

 **Under-determined**

$$N \leq r < M$$

 **Mixed-determined**

$$r < N \quad \text{and} \quad r < M$$

Inverse problem types - Even-determined

$$M = N = r$$

Generalized inverse = normal inverse

$$\mathbf{R}^M = \mathbf{R}^D = \mathbf{I}$$

Inverse problem types - Over-determined

$$N > r = M$$

Generalized inverse = least-squares inverse

$$\mathbf{G}^\dagger = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T = \mathbf{V}_r \boldsymbol{\Sigma}_r^{-2} \mathbf{V}_r^T \mathbf{V}_r \boldsymbol{\Sigma}_r \mathbf{U}_r^T = \mathbf{V}_r \boldsymbol{\Sigma}_r^{-1} \mathbf{U}_r^T$$

$$\mathbf{R}^M = \mathbf{I} \quad , \quad \mathbf{R}^D \neq \mathbf{I}$$

Inverse problem types - Under-determined

$$N = r < M$$

Generalized inverse = minimum-norm inverse

$$\mathbf{G}^\dagger = \mathbf{G}^T (\mathbf{G} \mathbf{G}^T)^{-1} = \mathbf{V}_r \boldsymbol{\Sigma}_r \mathbf{U}_r^T \mathbf{U}_r \boldsymbol{\Sigma}_r^{-2} \mathbf{U}_r^T = \mathbf{V}_r \boldsymbol{\Sigma}_r^{-1} \mathbf{U}_r^T$$

$$\mathbf{R}^M \neq \mathbf{I} \quad , \quad \mathbf{R}^D = \mathbf{I}$$

Inverse problem types - Mixed-determined

$$r < M \quad \text{and} \quad r < M$$

The generalized inverse

$$\mathbf{G}^\dagger = \mathbf{V}_r \mathbf{\Sigma}_r^{-1} \mathbf{U}_r^T$$

handles both over-determined and underdetermined model parts.

$$\mathbf{R}^M \neq \mathbf{I} \quad , \quad \mathbf{R}^D \neq \mathbf{I}$$

The problem of small eigenvalues

Generalized inverse $\mathbf{G}^\dagger = \mathbf{V}_r \mathbf{\Sigma}_r^{-1} \mathbf{U}_r^T$ to invert noisy data $\mathbf{G}\mathbf{m} + \mathbf{n}$

$$\mathbf{G}^\dagger(\mathbf{G}\mathbf{m} + \mathbf{n}) = \mathbf{G}^\dagger\mathbf{G}\mathbf{m} + \mathbf{G}^\dagger\mathbf{n} = \mathbf{R}^M + \sum_{i=1}^r \frac{\mathbf{u}_i^T \mathbf{n}}{\sigma_i} \mathbf{v}_i$$

! Problem

Small eigenvalues amplify noise in the data!

Moderate Example

$$\mathbf{G} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1.1 & 0 & 0 \\ 0 & 0 & 1 & 0.5 \\ 0 & 0 & 0.5 & 1 \end{pmatrix}$$

$$\Rightarrow \sigma_i = [2.05, 1.5, 0.5, 0.05]$$

$$\mathbf{m} = \text{pinv}(\mathbf{G}) * (\mathbf{d} + \mathbf{n})$$

$$\mathbf{m} = [10, 11, 12, 13], \epsilon = 0.2$$

$$\Rightarrow \mathbf{d} = [21.0, 22.1, 18.5, 19.0]$$

$$\mathbf{m}^1 = [4.45, 16.24, 11.62, 13.37]$$

$$\mathbf{m}^2 = [9.92, 10.92, 12.36, 12.59]$$

$$\mathbf{m}^3 = [15.11, 6.2, 11.66, 13.41]$$

$$\mathbf{m}^4 = [4.11, 16.76, 11.74, 13.05]$$

$$\mathbf{m}^5 = [4.4, 16.48, 11.55, 13.19]$$

Even a small noise level let the solution explode.

Solution: drop small singular values

$$\mathbf{G} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1.1 & 0 & 0 \\ 0 & 0 & 1 & 0.5 \\ 0 & 0 & 0.5 & 1 \end{pmatrix}$$

$$\Rightarrow \sigma_i = [2.05, 1.5, 0.5, 0.05]$$

$$\Rightarrow \mathbf{m} = \text{pinv}(\mathbf{G}, \text{rtol}=0.1) * (\mathbf{d} + \mathbf{n})$$

$$\mathbf{m} = [10, 11, 12, 13], \epsilon = 0.2$$

$$\Rightarrow \mathbf{d} = [21.0, 22.1, 18.5, 19.0]$$

$$\mathbf{m}^1 = [10.36, 10.89, 12.07, 13.2]$$

$$\mathbf{m}^2 = [10.29, 10.82, 11.67, 13.08]$$

$$\mathbf{m}^3 = [10.32, 10.84, 11.97, 13.16]$$

$$\mathbf{m}^4 = [10.25, 10.78, 11.81, 13.04]$$

$$\mathbf{m}^3 = [10.29, 10.82, 11.71, 13.13]$$

The solution is much less noise-depending!

Truncated singular value (TSVD) method

- Look at the singular value spectrum
- Choose a maximum number p or `rtol` and compute (e.g. by `pinv`)

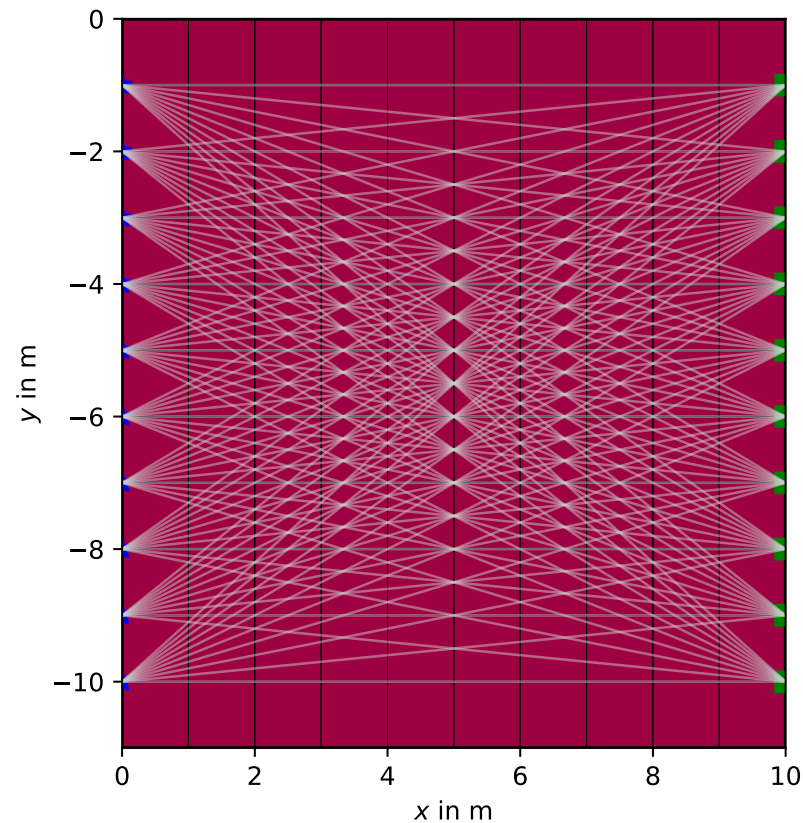
$$\mathbf{G}_p^\dagger = \mathbf{V}_p \mathbf{\Sigma}_p^{-1} \mathbf{U}_p^T$$

- How to choose p ? Trade-off between resolution and artifacts.

i Discrepancy principle (free after Occam)

Look at data fit and choose p such that the data can be fitted within the error.

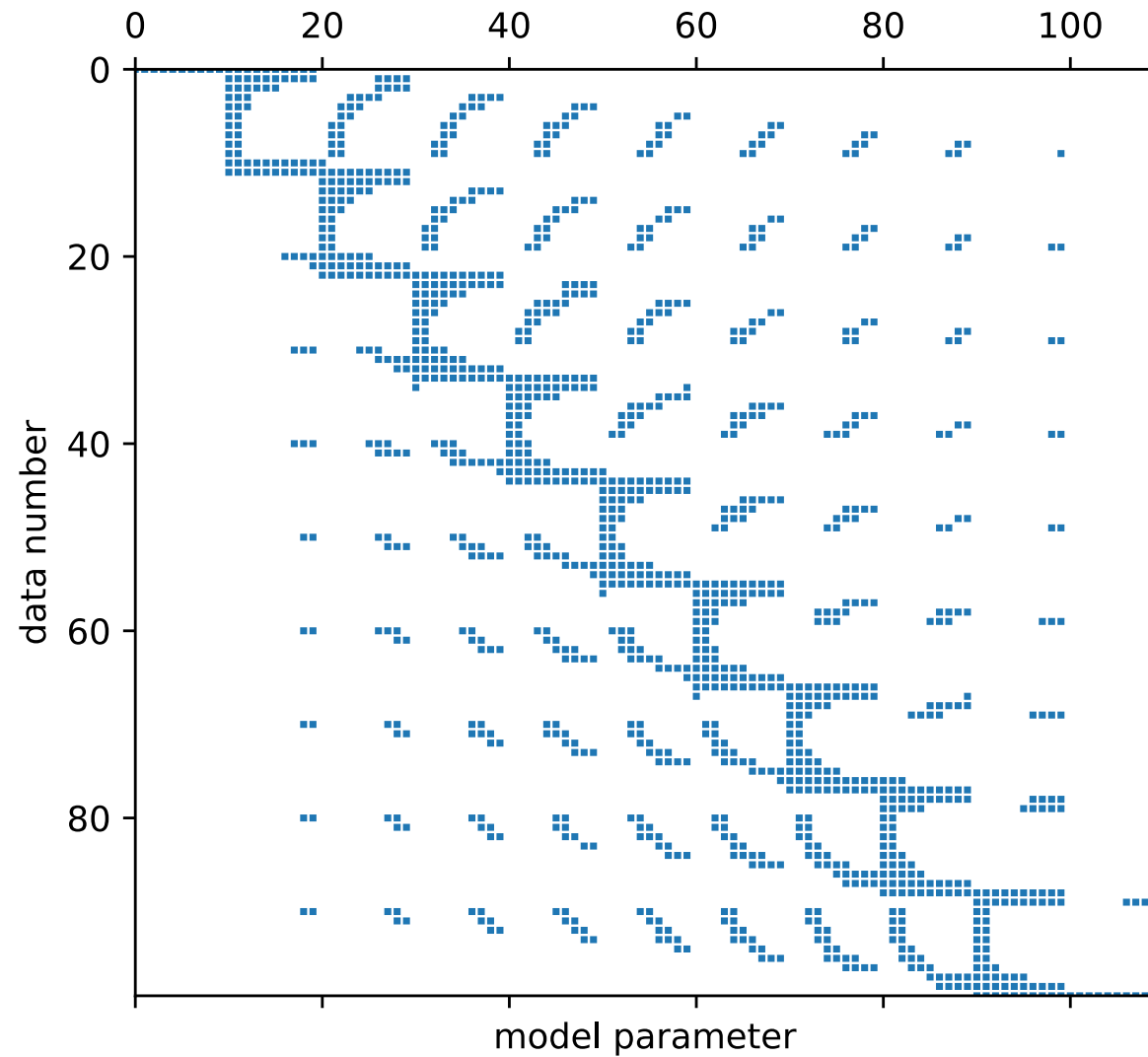
Geophysical example



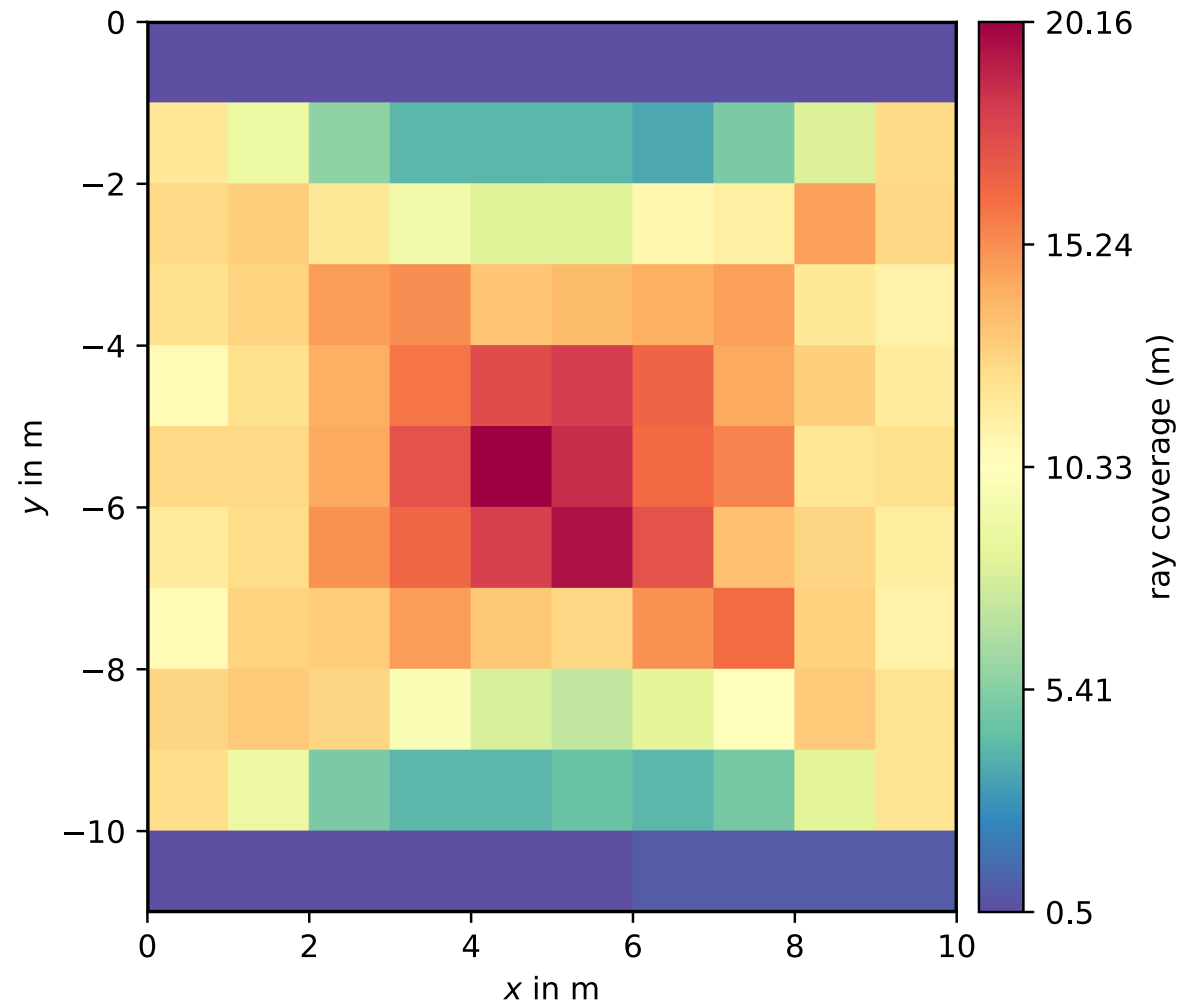
Seismic crosshole tomography

- grid with 1m spacing (11x10 cells)
- two boreholes: shots left, geophones right, fully connected (10x10 data)
- straight ray paths (x-ray, small contrasts)

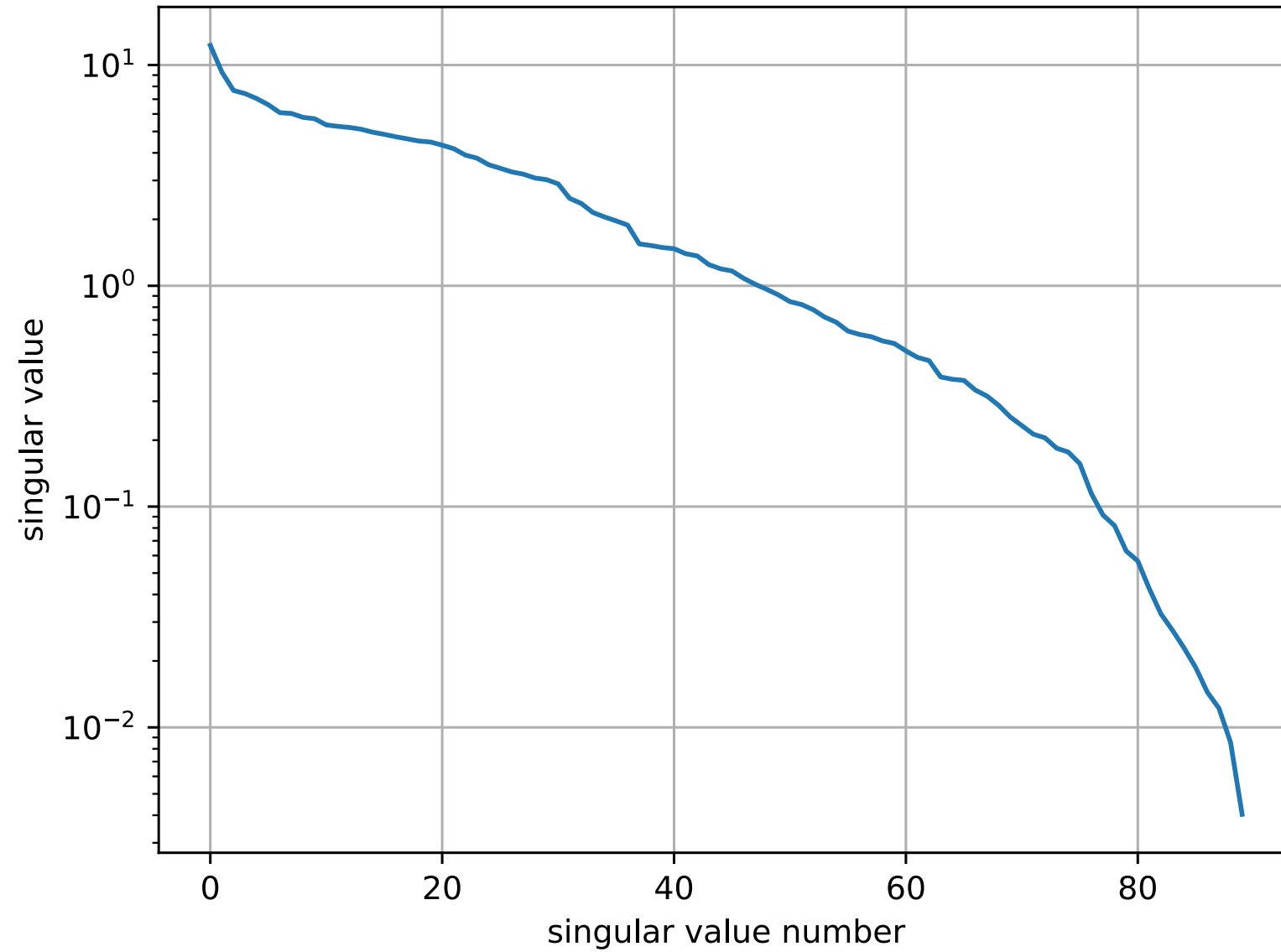
Way matrix and coverage



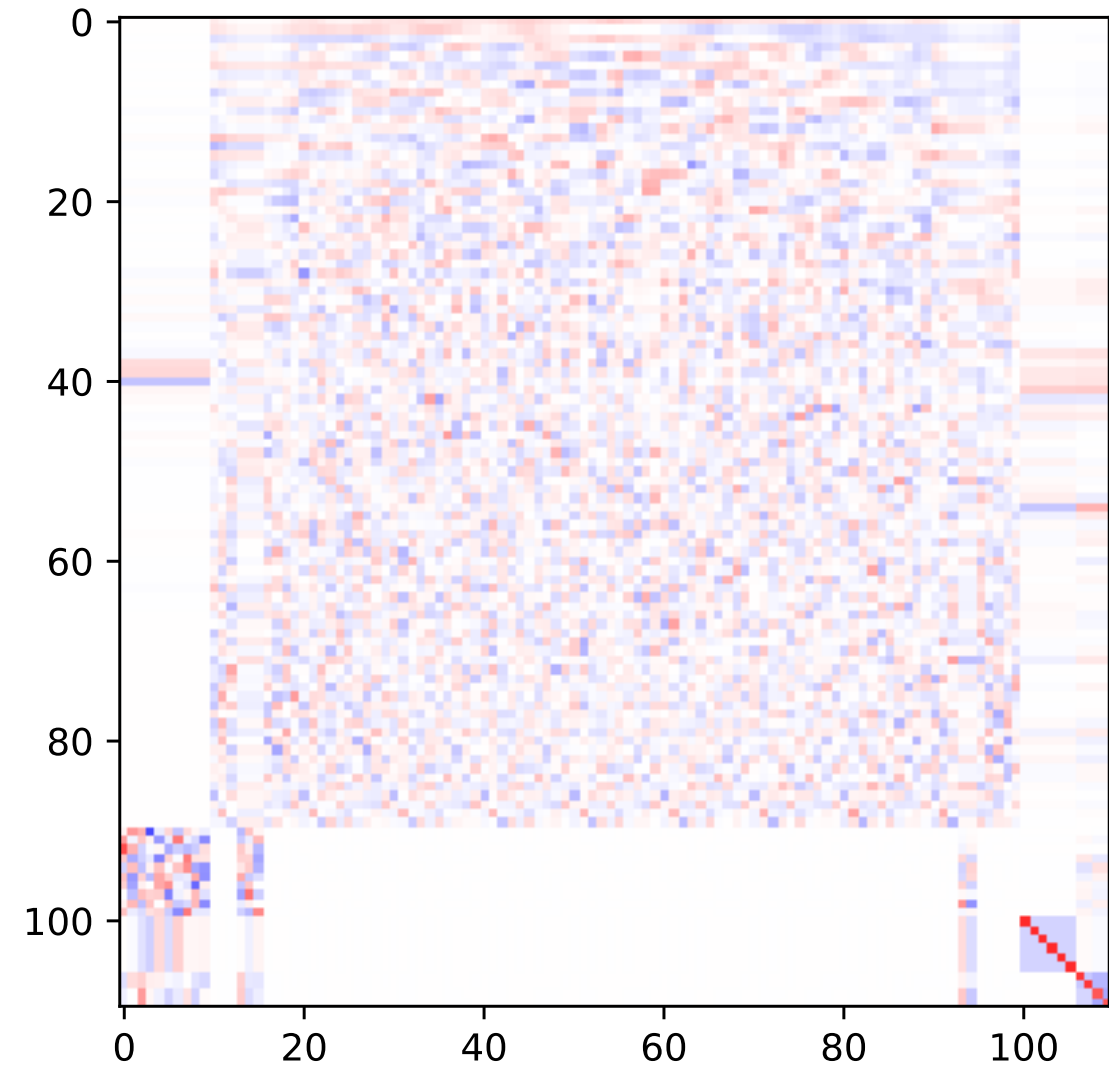
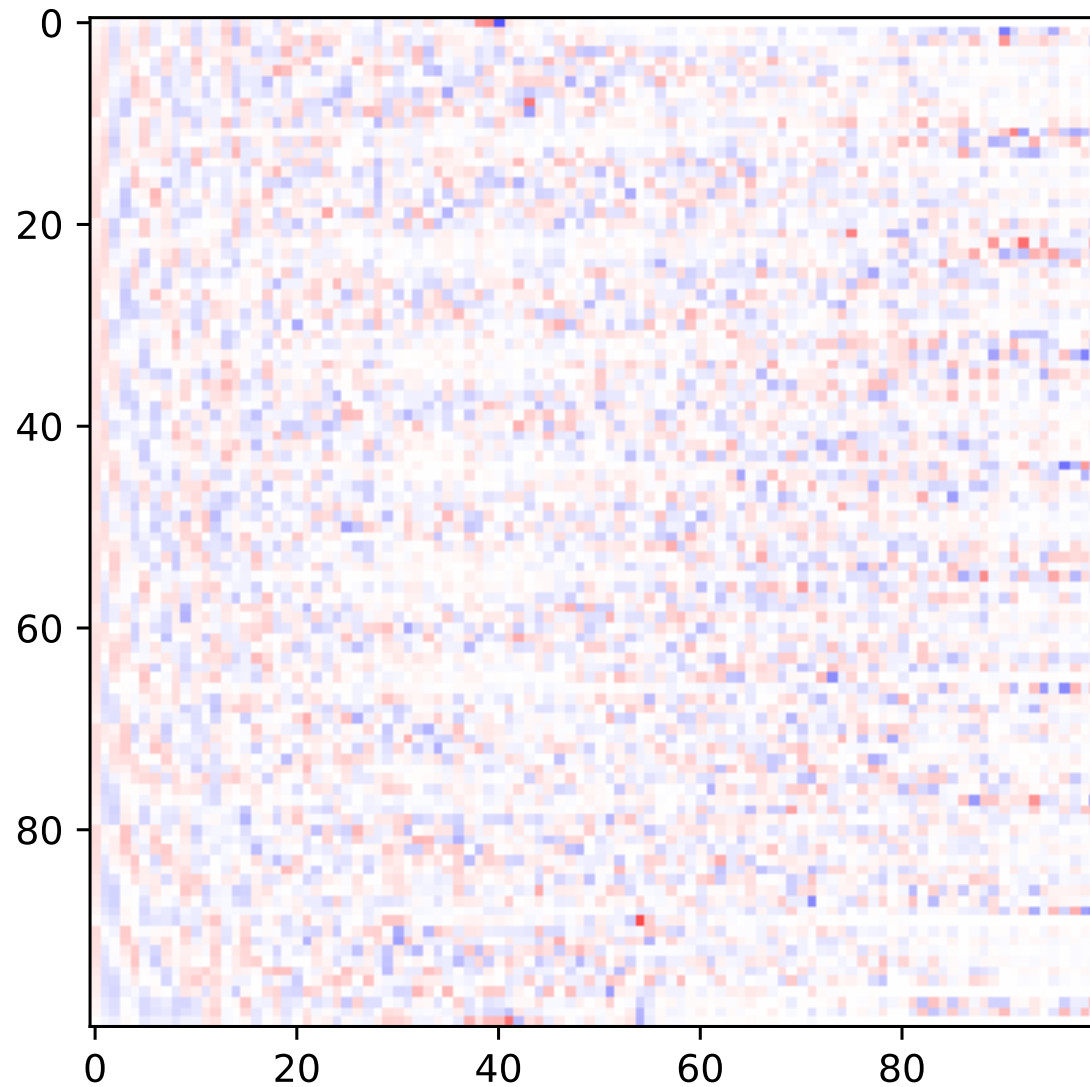
Coverage: data-sum of Jacobian



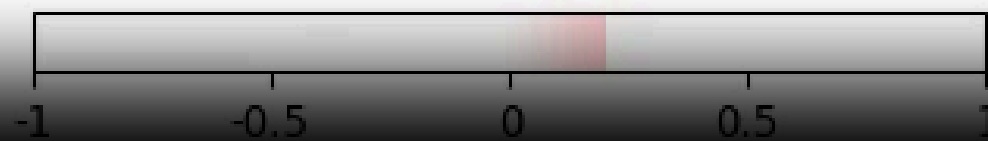
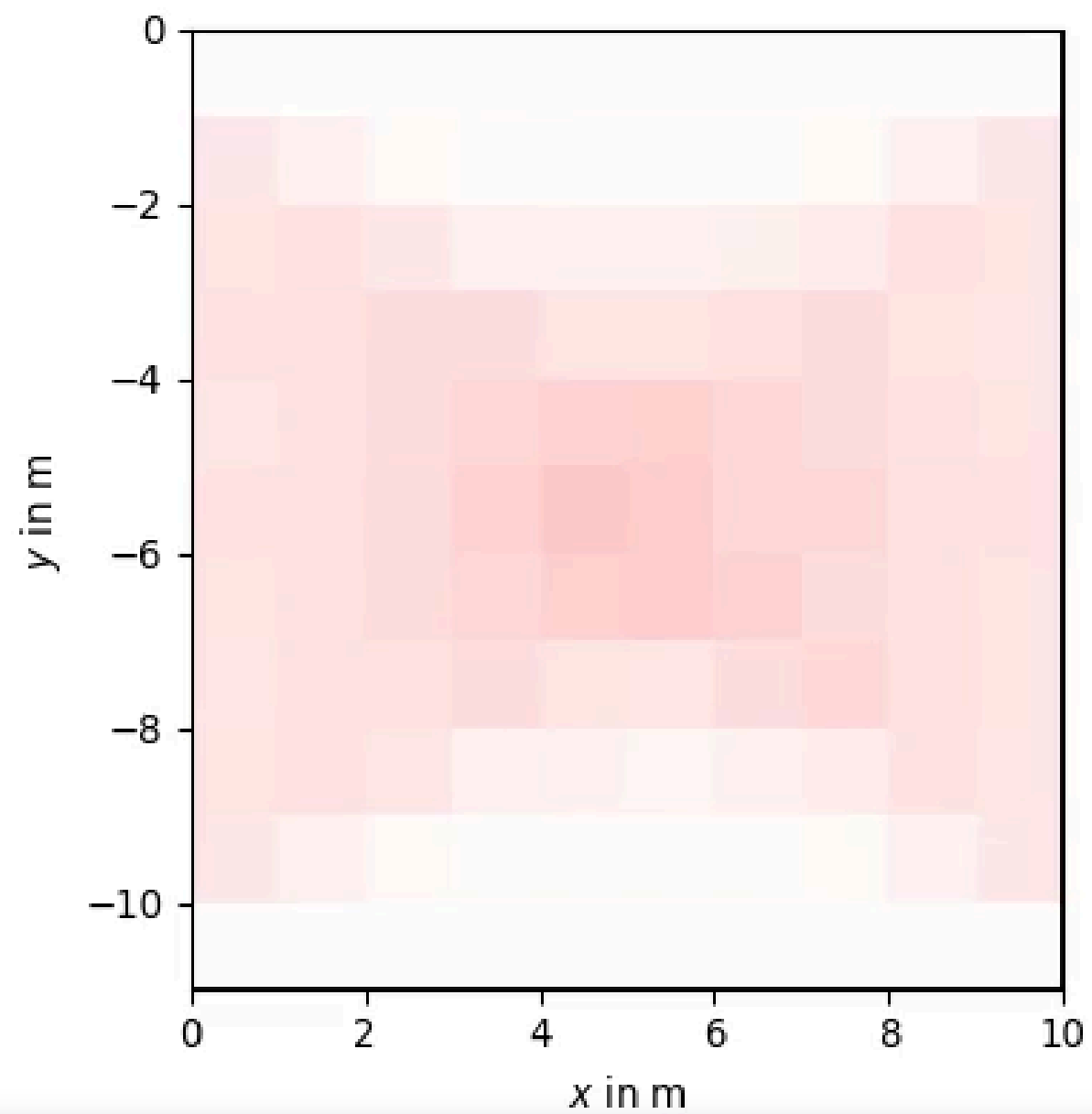
Singular value spectrum



Data and model eigenvectors



Model eigenvectors



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Data and model resolution matrix

